

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Aqualine™ Matrix K

**Recommended Use** Laboratory chemicals.  
**Uses advised against** All other uses

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | K/2300R/08, K/2300R/15                                                                      |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Flammable liquids

Category 3

#### Health hazards

Acute Oral Toxicity  
 Acute Inhalation Toxicity - Vapors  
 Skin Corrosion/Irritation  
 Serious Eye Damage/Eye Irritation  
 Carcinogenicity  
 Reproductive Toxicity  
 Specific target organ toxicity - (single exposure)  
 Specific target organ toxicity - (repeated exposure)

Category 4  
 Category 3  
 Category 2  
 Category 2  
 Category 2  
 Category 2  
 Category 3  
 Category 1

#### Environmental hazards

Chronic aquatic toxicity

Category 4

### Label Elements



Signal Word

Danger

**Hazard Statements**

H226 - Flammable liquid and vapor  
 H302 - Harmful if swallowed  
 H319 - Causes serious eye irritation  
 H315 - Causes skin irritation  
 H331 - Toxic if inhaled  
 H336 - May cause drowsiness or dizziness  
 H351 - Suspected of causing cancer  
 H361 - Suspected of damaging fertility or the unborn child  
 H372 - Causes damage to organs through prolonged or repeated exposure  
 H413 - May cause long lasting harmful effects to aquatic life

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling  
 P201 - Obtain special instructions before use  
 P202 - Do not handle until all safety precautions have been read and understood  
 P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
 P233 - Keep container tightly closed  
 P240 - Ground and bond container and receiving equipment  
 P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
 P242 - Use non-sparking tools  
 P243 - Take action to prevent static discharges  
 P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
 P270 - Do not eat, drink or smoke when using this product  
 P271 - Use only outdoors or in a well-ventilated area  
 P280 - Wear eye protection/ face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
 P330 - Rinse mouth  
 P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
 P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
 P311 - Call a POISON CENTER or doctor  
 P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
 P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
 P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component       | CAS No   | Weight % |
|-----------------|----------|----------|
| Chloroform      | 67-66-3  | 50-70    |
| 2-Butoxyethanol | 111-76-2 | 10 - 20  |
| Ethyl alcohol   | 64-17-5  | 5 - 10   |

|                 |          |        |
|-----------------|----------|--------|
| Ethylene glycol | 107-21-1 | 5 - 10 |
|-----------------|----------|--------|

## Section 4 - First Aid Measures

### Description of first aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Inhalation may cause anesthesia. Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                             |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                     |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. If skin irritation persists, call a physician.                                                                                                                                                                           |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Drink 1 or 2 glasses of water. Call a physician or poison control center immediately.                                                                                                                                                                                                                  |
| <b>Self-Protection of the First Aider</b>  | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                 |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                           |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. May cause cardiac arrhythmia. Symptoms of overexposure are dizziness, headache, tiredness, nausea, unconsciousness, cessation of breathing: May cause decreases in blood pressure and other cardiac effects                                                                                           |
| <b>Notes to Physician</b>                  | Treat symptomatically. Signs of overdose include stupor and respiratory depression. Symptoms may be delayed.                                                                                                                                                                                                                   |

## Section 5 - Fire Fighting Measures

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Flammable. Containers may explode when heated. Vapors may travel to source of ignition and flash back. Vapors may form explosive mixtures with air.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Phosgene, Hydrogen chloride gas.

### **Special protective equipment and precautions for fire fighters**

Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

**Emergency procedures**

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame. Protect from direct sunlight. Keep away from acids.

**Incompatible Materials**

Strong oxidizing agents. Alkali metals. Aluminium. Acetone.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component       | New Zealand WEL                                  | Australia                               | ACGIH TLV   | The United Kingdom                                                                      |
|-----------------|--------------------------------------------------|-----------------------------------------|-------------|-----------------------------------------------------------------------------------------|
| Chloroform      | TWA: 2 ppm<br>TWA: 9.9 mg/m <sup>3</sup><br>Skin | TWA: 2 ppm<br>TWA: 10 mg/m <sup>3</sup> | TWA: 10 ppm | TWA: 2 ppm<br>TWA: 9.9 mg/m <sup>3</sup><br>STEL: 6 ppm<br>STEL: 29.7 mg/m <sup>3</sup> |
| 2-Butoxyethanol | TWA: 25 ppm                                      | STEL: 50 ppm                            | TWA: 20 ppm | STEL: 50 ppm 15 min                                                                     |

|                 |                                                   |                                                                                                                      |                                                           |                                                                                                                                                                                                |
|-----------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | TWA: 121 mg/m <sup>3</sup><br>Skin                | STEL: 242 mg/m <sup>3</sup><br>TWA: 20 ppm<br>TWA: 96.9 mg/m <sup>3</sup>                                            |                                                           | STEL: 246 mg/m <sup>3</sup> 15 min<br>TWA: 25 ppm 8 hr<br>TWA: 123 mg/m <sup>3</sup> 8 hr<br>Skin                                                                                              |
| Ethyl alcohol   | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>      | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>                                                                         | STEL: 1000 ppm                                            | TWA: 1000 ppm TWA; 1920 mg/m <sup>3</sup> TWA<br>WEL - STEL: 3000 ppm<br>STEL; 5760 mg/m <sup>3</sup> STEL                                                                                     |
| Ethylene glycol | Ceiling: 50 ppm<br>Ceiling: 127 mg/m <sup>3</sup> | STEL: 40 ppm<br>STEL: 104 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup><br>TWA: 20 ppm<br>TWA: 52 mg/m <sup>3</sup> | TWA: 25 ppm<br>STEL: 50 ppm<br>STEL: 10 mg/m <sup>3</sup> | STEL: 40 ppm 15 min<br>STEL: 104 mg/m <sup>3</sup> 15 min<br>STEL: 30 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr<br>TWA: 20 ppm 8 hr<br>TWA: 52 mg/m <sup>3</sup> 8 hr<br>Skin |

**Biological limit values**

**UK** - Biological Monitoring Guidance Values provided by the UK's Health and Safety Executive (HSE) Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended) and EH40/2005.

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

| Component       | New Zealand | Australia | ACGIH - Biological Exposure Indices                                                                          | United Kingdom                                              |
|-----------------|-------------|-----------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 2-Butoxyethanol |             |           | 200 mg/g creatinine<br>Medium: urine<br>Time: end of shift<br>Determinant: Butoxyacetic acid with hydrolysis | Butoxyacetic acid: 240 mmol/mol creatinine urine post shift |

**Appropriate engineering controls****Engineering Measures**

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

**Eye Protection** Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                 |
|----------------|-------------------|-----------------|-----------------|--------------------------------------------------------------------------------|
| Viton (R).     | > 480 minutes     | -               | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Neoprene       | < 25 minutes      | 0.45 mm         |                 |                                                                                |
| Butyl rubber   | < 15 minutes      | 0.35 mm         |                 |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 (or AUS/NZ equivalent)

|                                        |                                                                                                                                                     |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                              |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system.                                                     |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Clear Colorless          |                                          |
| <b>Odor</b>                                    | Characteristic           |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | Flammable                | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | > 36.1 °C / > 97 °F      | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Partially soluble        |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Chloroform                                     | 2                        |                                          |
| 2-Butoxyethanol                                | 0.81                     |                                          |
| Ethyl alcohol                                  | -0.32                    |                                          |
| Ethylene glycol                                | -1.36                    |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 1.22                     |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | > 1.0                    | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### Other information

|                             |                                        |
|-----------------------------|----------------------------------------|
| <b>Explosive Properties</b> | explosive air/vapour mixtures possible |
|-----------------------------|----------------------------------------|

## Section 10 - Stability and Reactivity

|                                         |                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | Sensitivity to light; Moisture sensitive                                                          |
| <b>Stability</b>                        | Stable under normal conditions. UNSTABLE (REACTIVE) UPON DEPLETION OF INHIBITOR. Light sensitive. |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                          |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                          |

|                                         |                                                                                                                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazardous Reactions</b>              | No information available.                                                                                                                                                |
| <b>Conditions to Avoid</b>              | Incompatible products, Heat, flames and sparks, Excess heat, Exposure to light, Protect from moisture, Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Alkali metals, Aluminium, Acetone.                                                                                                              |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Phosgene. Hydrogen chloride gas.                                                                                |

## Section 11 - Toxicological Information

### Acute Effects

### Information on likely routes of exposure

#### Product Information

|                   |                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Harmful by inhalation. May cause irritation of respiratory tract.                                     |
| <b>Eyes</b>       | May cause irritation.                                                                                 |
| <b>Skin</b>       | Harmful in contact with skin. May cause irritation.                                                   |
| <b>Ingestion</b>  | Harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. |

### Numerical measures of toxicity

|                            |            |
|----------------------------|------------|
| <b>(a) acute toxicity;</b> |            |
| <b>Oral</b>                | Category 4 |
| <b>Dermal</b>              | Category 4 |
| <b>Inhalation</b>          | Category 3 |

### Toxicology data for the components

| Component       | LD50 Oral                                                                      | LD50 Dermal                                | LC50 Inhalation                                                   |
|-----------------|--------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|
| Chloroform      | LD50 = 908 mg/kg (rat)<br>LD50 = 695 mg/kg ( Rat )<br>LD50 = 450 mg/kg ( Rat ) | LD50 > 20 g/kg ( Rabbit )                  | LC50 = 10.5 mg/L ( Rat ) 4 h                                      |
| 2-Butoxyethanol | 1746 mg/kg ( Rat )                                                             | LD50 > 2000 mg/kg (Guinea pig)<br>OCED 402 | LC50 = 450 ppm ( Rat ) 4 h<br>LC50 = 486 ppm ( Rat ) 4 h          |
| Ethyl alcohol   | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse )                   |                                            | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |
| Ethylene glycol | LD50 = 4700 mg/kg ( Rat )                                                      | LD50 = 10600 mg/kg ( Rat )                 | LC50 > 2.5 mg/L ( Rat ) 6 h                                       |

| Component       | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|-----------------|-----------------------|-------------------------|-----------------------------|
| 2-Butoxyethanol | ATE = 1200 mg/kg bw   | -                       | ATE = 3 mg/L (vapour)       |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**  
**Respiratory** No data available  
**Skin** No data available

| Component                               | Test method                         | Test species | Study result      |
|-----------------------------------------|-------------------------------------|--------------|-------------------|
| 2-Butoxyethanol<br>111-76-2 ( 10 - 20 ) | Guinea Pig Maximisation Test (GPMT) | guinea pig   | - non-sensitising |
| Ethyl alcohol<br>64-17-5 ( 5 - 10 )     | Mouse Ear Swelling Test (MEST)      | mouse        | non-sensitising   |
|                                         | OECD Test Guideline 429 Local       | mouse        | non-sensitising   |

|  |                  |  |  |
|--|------------------|--|--|
|  | Lymph Node Assay |  |  |
|--|------------------|--|--|

(e) germ cell mutagenicity; No data available

| Component                           | Test method                                   | Test species          | Study result |
|-------------------------------------|-----------------------------------------------|-----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 5 - 10 ) | AMES test<br>OECD Test Guideline 471          | in vitro<br>Bacteria  | negative     |
|                                     | Gene cell mutation<br>OECD Test Guideline 476 | in vitro<br>Mammalian | negative     |

(f) carcinogenicity; Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component  | New Zealand                | Australia            | New South Wales | Western Australia | IARC     | EU | UK | Germany |
|------------|----------------------------|----------------------|-----------------|-------------------|----------|----|----|---------|
| Chloroform | Suspected human carcinogen | suspected carcinogen |                 |                   | Group 2B |    |    |         |

(g) reproductive toxicity; Category 2

| Component                           | Test method             | Test species / Duration   | Study result          |
|-------------------------------------|-------------------------|---------------------------|-----------------------|
| Ethyl alcohol<br>64-17-5 ( 5 - 10 ) | OECD Test Guideline 416 | Oral / mouse 2 Generation | NOAEL = 13.8 g/kg/day |
|                                     | OECD Test Guideline 414 | Inhalation / Rat          | NOAEC = 16000 ppm     |

**Developmental Effects** Developmental effects have occurred in experimental animals

(h) STOT-single exposure; Category 3

**Results / Target organs** Central nervous system (CNS)

(i) STOT-repeated exposure; Category 1

**Study result** LOAEL = 15 mg/kg bw/day  
NOAEC = 25 mg/m<sup>3</sup>  
**Target Organs** Liver, Kidney, Nasal Cavities.

(j) aspiration hazard; Based on available data, the classification criteria are not met

#### Symptoms / effects, both acute and delayed

Symptoms of overexposure are dizziness, headache, tiredness, nausea, unconsciousness, cessation of breathing. May cause decreases in blood pressure and other cardiac effects.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component  | Freshwater Fish                                                                                                                                                                                  | Water Flea           | Freshwater Algae    | Microtox                                                                                                                                                     |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chloroform | LC50: = 300 mg/L, 96h static (Poecilia reticulata)<br>LC50: = 18 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: = 18 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: = 71 mg/L, 96h | EC50 = 28.9 mg/L/48h | EC50 = 560 mg/L/48h | Photobacterium phosphoreum: EC50 = 520 mg/L/5 min<br>Photobacterium phosphoreum: EC50 = 670 mg/L/15 min<br>Photobacterium phosphoreum: EC50 = 670 mg/L/30min |



|                 |                                                                                                                                                                                                                                                                                                                                                                                 |                                                                               |                                                                         |                                                                                                                       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|                 | flow-through<br>(Pimephales promelas)                                                                                                                                                                                                                                                                                                                                           |                                                                               |                                                                         |                                                                                                                       |
| 2-Butoxyethanol | 1490 mg/L LC50 96 h<br>2950 mg/L LC50 96 h                                                                                                                                                                                                                                                                                                                                      | 1550 mg/L EC50 48 hr<br>>1000 mg/L EC50 48 h<br>1698 - 1940 mg/L EC50<br>24 h | 1840 mg/L EC50 72 hr                                                    |                                                                                                                       |
| Ethyl alcohol   | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/L/96h                                                                                                                                                                                                                                                                                                                | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h                                 | EC50 (72h) = 275 mg/L<br>(Chlorella vulgaris)                           | Photobacterium<br>phosphoreum:EC50 =<br>34634 mg/L/30 min<br>Photobacterium<br>phosphoreum:EC50 =<br>35470 mg/L/5 min |
| Ethylene glycol | LC50: 14 - 18 mL/L,<br>96h static<br>(Oncorhynchus mykiss)<br>LC50: = 27540 mg/L,<br>96h static (Lepomis<br>macrochirus)<br>LC50: = 40761 mg/L,<br>96h static<br>(Oncorhynchus mykiss)<br>LC50: 40000 - 60000<br>mg/L, 96h static<br>(Pimephales promelas)<br>LC50: = 16000 mg/L,<br>96h static (Poecilia<br>reticulata)<br>LC50: = 41000 mg/L,<br>96h (Oncorhynchus<br>mykiss) | EC50: = 46300 mg/L,<br>48h (Daphnia magna)                                    | EC50: 6500 - 13000<br>mg/L, 96h<br>(Pseudokirchneriella<br>subcapitata) | EC50 = 10000 mg/L 16<br>h<br>EC50 = 620 mg/L 30<br>min<br>EC50 = 620.0 mg/L 30<br>min                                 |

**Terrestrial ecotoxicity**

| Component     | Earthworm                                                                                   | Avian | Honeybees |
|---------------|---------------------------------------------------------------------------------------------|-------|-----------|
| Ethyl alcohol | Acute toxicity: LC50 0.1 - 1<br>mg/cm <sup>2</sup> (Eisenia foetida, 48 h,<br>filter paper) |       |           |

**Persistence and Degradability** No data is available on the product itself

| Component                               | Degradability       |
|-----------------------------------------|---------------------|
| 2-Butoxyethanol<br>111-76-2 ( 10 - 20 ) | 90% (28d) OECD 301B |
| Ethyl alcohol<br>64-17-5 ( 5 - 10 )     | OECD 301E = 94%     |

**Bioaccumulative Potential** There is no data for this product

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Chloroform      | 2       | 1.4 - 13 dimensionless        |
| 2-Butoxyethanol | 0.81    | No data available             |
| Ethyl alcohol   | -0.32   | No data available             |
| Ethylene glycol | -1.36   | No data available             |

**Mobility** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. .

**Other adverse effects**

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

#### Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### Other Information

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

## Section 14 - Transport Information

| Component                           | Hazchem Code |
|-------------------------------------|--------------|
| Chloroform<br>67-66-3 ( 50-70 )     | 2Z           |
| Ethyl alcohol<br>64-17-5 ( 5 - 10 ) | 2YE<br>2Y    |

### NZS 5433:2020

|                         |                                 |
|-------------------------|---------------------------------|
| UN-No                   | UN1992                          |
| Proper Shipping Name    | Flammable liquid, toxic, n.o.s. |
| Technical Shipping Name | Contains ethanol, chloroform    |
| Hazard Class            | 3                               |
| Subsidiary Hazard Class | 6.1                             |
| Packing Group           | III                             |

### IATA

|                         |                                 |
|-------------------------|---------------------------------|
| UN-No                   | UN1992                          |
| Proper Shipping Name    | Flammable liquid, toxic, n.o.s. |
| Technical Shipping Name | Contains ethanol, chloroform    |
| Hazard Class            | 3                               |
| Subsidiary Hazard Class | 6.1                             |
| Packing Group           | III                             |

### IMDG/IMO

|                         |                                 |
|-------------------------|---------------------------------|
| UN-No                   | UN1992                          |
| Proper Shipping Name    | Flammable liquid, toxic, n.o.s. |
| Technical Shipping Name | Contains ethanol, chloroform    |
| Hazard Class            | 3                               |
| Subsidiary Hazard Class | 6.1                             |
| Packing Group           | III                             |

**Environmental hazards** No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

Additional information

None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

| Component  | New Zealand                |
|------------|----------------------------|
| Chloroform | Suspected human carcinogen |

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Authorisation/Restrictions according to EU REACH

| Component       | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                        | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Chloroform      | -                                                                   | Use restricted. See item 32. (see <a href="http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32006R1907:EN:NOT">http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32006R1907:EN:NOT</a> for restriction details) | -                                                                                                     |
| 2-Butoxyethanol | -                                                                   | Use restricted. See item 75. (see link for restriction details)                                                                                                                                                                      | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No   | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-----------------|----------|-------|------|-----------|--------|-----|----------|-------|------|
| Chloroform      | 67-66-3  | X     | X    | 200-663-8 | -      | -   | X        | X     | X    |
| 2-Butoxyethanol | 111-76-2 | X     | X    | 203-905-0 | -      | -   | KE-04134 | X     | X    |
| Ethyl alcohol   | 64-17-5  | X     | X    | -         | -      | -   | KE-13217 | X     | X    |
| Ethylene glycol | 107-21-1 | X     | X    | 203-473-3 | -      | -   | KE-13169 | X     | X    |

| Component | CAS No | TSCA | TSCA Inventory | DSL | NDSL | PICCS | ISHL | ENCS |
|-----------|--------|------|----------------|-----|------|-------|------|------|
|-----------|--------|------|----------------|-----|------|-------|------|------|

|                 |          |   | notification -<br>Active-Inactive |   |   |   |   |   |
|-----------------|----------|---|-----------------------------------|---|---|---|---|---|
| Chloroform      | 67-66-3  | X | ACTIVE                            | X | - | X | X | X |
| 2-Butoxyethanol | 111-76-2 | X | ACTIVE                            | X | - | X | X | X |
| Ethyl alcohol   | 64-17-5  | X | ACTIVE                            | X | - | X | X | X |
| Ethylene glycol | 107-21-1 | X | ACTIVE                            | X | - | X | X | X |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

NZIoC - New Zealand Inventory of Chemicals

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

IECSC - Chinese Inventory of Existing Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

NZS 5433:2020 - Transport of Dangerous Goods on Land

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

WEL - Workplace Exposure Limit

DNEL - Derived No Effect Level

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

VOC - (Volatile Organic Compound)

AICS - Australian Inventory of Chemical Substances

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

CAS - Chemical Abstracts Service

ACGIH - American Conference of Governmental Industrial Hygienists

PNEC - Predicted No Effect Concentration

OECD - Organisation for Economic Co-operation and Development

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail

LC50 - Lethal Concentration 50%

ATE - Acute Toxicity Estimate

RPE - Respiratory Protective Equipment

NOEC - No Observed Effect Concentration

BCF - Bioconcentration factor

PBT - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Revision Date** 13-Mar-2023

**Revision Summary** Not applicable

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

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date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**